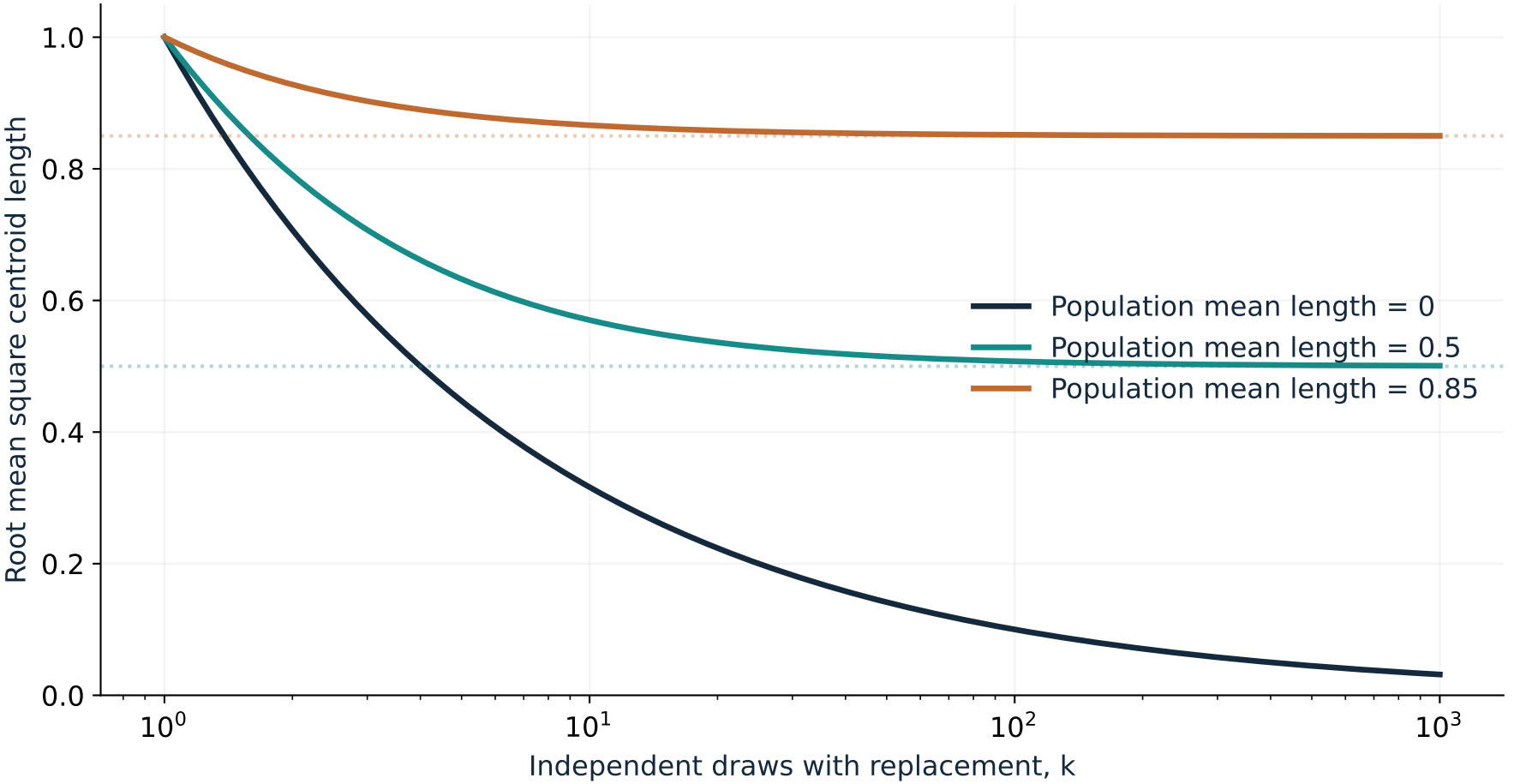
Averages approach the population mean

$$\sqrt{E\|\bar{X}_k\|^2} = \sqrt{\|\mu\|^2 + (E\|X\|^2 - \|\mu\|^2)/k}$$



Analytic unit-vector examples. These curves are RMS lengths, not mean lengths.
The three mean lengths are illustrative; none is a measured Unicode subset.